Discrete Random Variables

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Question 1

0/1 pt 3 19

Your favorite basketball player is a 85% free throw shooter. Find the probability (as a percent) that he does NOT make his next free throw shot.

_______ %

Write your answer as a percent.

Question Help: Video

Question 2

0/1 pt 3 19

A bag contains 7 Cherry Starbursts and 23 other flavored Starburts. 12 Starbursts are chosen randomly without replacement.
Find the probability that 2 of the Starbursts drawn are cherry.

Question Help: Video Read

Question 3

0/1 pt 3 19

An accounting office has 6 incoming telephone lines. The probability distribution of the number of busy lines, X, is as follows:

x_i	0	1	2	3	4	5	6	Total
$P(X = x_i)$	0.02	0.15	0.2	0.31	0.17	0.11	0.04	1

1. Use the random variable notation to express symbolically each of the following:

   [-] The probability of an event in which the number of busy lines is exactly 4 is equal to 0.17.

   [-] The probability of an event in which the number of busy lines is exactly 4.

   [-] An event in which the number of busy lines is exactly 1.

2. Use the above probability distribution table to find the following:
   a. $P(X = 0) =$ ______ (Round the answer to 2 decimals)
   b. $P(5 \leq X) =$ ______ (Round the answer to 2 decimals)
   c. $P(X \leq 5) =$ ______ (Round the answer to 2 decimals)

Question Help: Video

● **Question 4** 0/1 pt 3 19

The number of pets per household in a certain neigborhood varies from 0 to 6. The probability distribution of the number of pets owned by a random household, X, is as follows:

x_i	0	1	2	3	4	5	6	Total
$P(X = x_i)$	0.05	0.13	0.2	0.33	0.16	0.12	0.01	1

1. Use the random variable notation to express symbolically each of the following:

   [-] An event in which the number of pets owned by a random household is exactly 6.

   [-] The probability that the number of pets owned by a random household is exactly 4 is equal to 0.16.

   [-] The probability that the number of pets owned by a random household is exactly 3.
2. Use the above probability distribution table to find the following:
   a. $P(2 < X < 4) =$ ______ (Round the answer to 2 decimals)
   b. $P(0 < X \leq 2) =$ ______ (Round the answer to 2 decimals)
   c. $P(X \geq 1) =$ ______ (Round the answer to 2 decimals)

Question Help: Video

Question 5 0/1 pt 3 19

A poll is given, showing 75% are in favor of a new building project.

If 7 people are chosen at random, what is the probability that exactly 3 of them favor the new building project?

[ ]

Question Help: Video

Question 6 0/1 pt 3 19

Assume that a procedure yields a binomial distribution with a trial repeated `n=5` times. Use some form of technology to find the probability distribution given the probability `p=0.625` of success on a single trial.

(Report answers accurate to 4 decimal places.)

k	***P*(*X* = *k*)**
0	
1	
2	
3	
4	
5	

Question Help: Video

Question 7

0/1 pt 3 19

A large fast-food restaurant is having a promotional game where game pieces can be found on various products. Customers can win food or cash prizes. According to the company, the probability of winning a prize (large or small) with any eligible purchase is 0.117.

Consider your next 29 purchases that produce a game piece. Calculate the following:

This is a binomial distribution. Round your answers to at least 4 decimal places.

a) What is the probability that you win 4 prizes?

b) What is the probability that you win more than 6 prizes?

c) What is the probability that you win between 2 and 5 (inclusive) prizes?

d) What is the probability that you win 1 prizes or fewer?

Question Help: Video Read

Question 8

0/1 pt 3 19

The probability of obtaining a defective 10-year old widget is 64%. For our purposes, the random variable will be the number of items that must be tested before finding the first defective 10-year old widget. Thus, this procedure yields a geometric distribution.

(Report answers accurate to 2 decimal places.)

k	***P(X = k)***
1	
2	
3	
4	
5	
6 or greater	

Question Help: Video

Question 9

0/1 pt 3 19

According to data from the American Medical Association, 10% of the population is left-handed. If 6 people are randomly selected, find the probability that ... (Round the answers to 4 decimal places.)

a) ... all of them are left-handed:

b) ... 3 of them are left-handed:

c) ... at most 5 are left-handed:

Question Help: Video

Question 10

0/1 pt 3 19

The number of magnitude 5 or larger earthquakes in Alaska follow a Poisson distribution with an average of 18 earthquakes per year. Round your answers to four decimals. In the following questions assume we are only talking about the magnitude 5 or larger earthquakes and the earthquakes are independent of one another. Round answers to at least 4 decimal places.

a) Find the probability of exactly 23 earthquakes in a particular year. ______

b) Find the probability of exactly one earthquake in one month. ______

c) Find the probability that there are no earthquakes in a month. ______

d) Find the probability that there are two earthquakes in a 3 month period.

______

Question 11

0/1 pt 3 19

Occasionally an airline will lose a bag. A small airline has found it loses an average of 6.3 bags each day. Find the probability that, on a given day,

(a) The airline loses exactly two bags.

(b) The airline loses fewer than fourteen bags.

(c) The airline loses more than thirteen bags.

*Round all answers to **four** decimal places.*

Question 12

0/1 pt 3 19

The Poisson Distribution
The Poisson Distribution is a **Discrete Probability Distribution** that is commonly applied when a series of trials/experiments occur over an interval. (The number of meteors per hour; hailstones per acre). Here, the average time/distance/etc. between each event must be known.

This type of distribution may be used if the following conditions apply:
Each event is independent.
The average rate is constant (events per interval).
Two events cannot occur at the same time.

Apply the Poisson Distribution to a scenario.
On average, a baking student accidentally drops three pieces of egg shell into the batter of every two cakes made.

The conditions of a Poisson Distribution are met:
The baking of a cake does not affect the outcome of any other cake. (independent events)
The average rate is constant: 3 pieces per 2 cakes `(rate = 3/2 = 1.5)`
Baking (and dropping eggshells) is a sequential process. (Two events cannot occur at the same time)

Use the scenario above to determine the expected value (`mu`) and selected probabilities below. You may wish to use the Poisson Distribution Calculator hosted by the University of Iowa's Department of Mathematical Sciences. Remember: the formatting of this calculator may vary slightly from what is used in class. (link: Poisson Distribution Calculator)

a. On average, how many egg shells do you expect to be in a single cake?
`mu = ` ______________ (decimal answers only)

b. If a single cake is bought, what is the probability of finding 0 egg shell pieces in it?
`P(X = 0) = ` ______________ (include four decimal places)

c. If a single cake is bought, what is the probability of finding *more than* two egg shell pieces in it?
`P(X > 2) = ` ______________ (include four decimal places)